








Statistics of Financial Markets

Seminar 5: Heavy tails and extreme value theory



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Acronyms used in this material

- **AI**: Artificial Intelligence
- **BET**: Bucharest Exchange Trading (index)
- **BVB**: Bursa de Valori București (the Bucharest Stock Exchange)
- **CDF**: Cumulative Distribution Function
- **CI**: Confidence Interval
- **EODHD**: EOD Historical Data (market-data provider)
- **ES**: Expected Shortfall
- **EVT**: Extreme Value Theory
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GEV**: Generalised Extreme Value (distribution)
- **GPD**: Generalised Pareto Distribution
- **ML**: Maximum Likelihood
- **OMV**: Österreichische Mineralölverwaltung — Austrian Mineral Oil Administration (OMV group)
- **POT**: Peaks Over Threshold
- **QQ**: Quantile–Quantile (plot)
- **SE**: Standard Error

Today's Question and Seminar Structure

- Question of the seminar
  - **how heavy are the tails of daily losses, and how large is the loss we should expect once in 1 000 days or once in 10 years?**
- Outputs to produce
  - the answers to the A exercises
  - the table or figure each B exercise asks for
  - a short argued answer to C1; the errors found in the AI answer of C2 and the corrected answer
  - the interpretation sentences each exercise asks for
- Nothing is handed in and nothing is graded: the seminar is for practice
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- Exercises, their types and their order: the Exercise Map (next slide)
- Contact: Prof. Daniel Traian Pele, danpele@ase.ro

Exercise Map

Task	Question	Type	Model
A1, A2	the Hill estimator and the Hill quantile from a few large losses	Solved, Proposed	A1
A3, A4	tail probabilities, VaR, ES and mean excess of a Pareto tail	Solved, Proposed	A3
A5, A6	GEV probabilities and return levels	Solved, Proposed	A5
A7, A8	VaR and ES from a POT fit, step by step	Solved, Proposed	A7
B1, B2	Hill plots and Hill estimates with standard errors for DAX, BET, Bitcoin and BVB stocks	Solved, Proposed	B1
B3, B4	POT for the BET and for Bitcoin: EVT vs historical vs Normal; the choice of the threshold	Solved, Proposed	B3
B5, B6	quarterly maxima of S&P 500 losses; an out-of-sample check of VaR 1%	Solved, Proposed	B5
C1, C2	has the tail of Bitcoin changed? what is wrong in an AI answer?	Proposed	B1, B3

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; the solution is discussed in class
- Part A: calculations on paper; Part B: real data in the notebook, each task ending with an interpretation question; Part C: open questions and the audit of an AI answer

Data Used

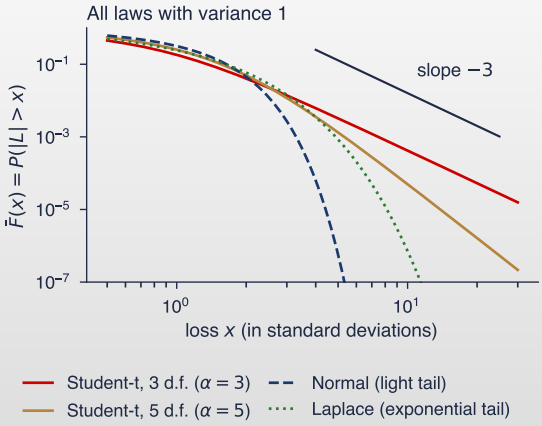
Series	Source	Price used	Period
S&P 500, DAX	EODHD	close	1990–2026
BET	EODHD	close	1997–2026
Bitcoin	EODHD	close, 7 days a week	2014–2026
OMV Petrom (SNP), Banca Transilvania (TLV)	EODHD	adjusted close	2010–2026

- Daily log returns in %: $r_t = 100\,(\ln P_t - \ln P_{t-1})$; daily losses $L_t = -r_t$; weekends and repeated holiday closes are dropped (except Bitcoin)
- TLV: the days 30 and 31 May 2016 are removed (a price adjustment applied one day late, Chapter 2)
- In the notebook: `losses('dax')`; no account or key is needed

Prerequisites for Today (1/19): Heavy Tails

- Losses $L = -r$ (in %); the **tail** of the losses: $\bar{F}(x) = P(L > x)$
  - $\bar{F}(x)$: the probability of a loss larger than x; it falls towards 0 as x grows
- **Heavy (power) tail** with **tail index** $\alpha > 0$: $\bar{F}(x) \approx Cx^{-\alpha}$ for large x
  - $C > 0$: a constant; the smaller α, the heavier the tail
  - doubling the loss divides its probability by 2^{α}
- **Light tail** (Normal, exponential)
  - $\bar{F}(x)$ falls faster than any power of x
- Moments: $E|L|^m < \infty$ only for $m < \alpha$
  - $E|L|^m$: the moment of order m; $\alpha \leq 2$: no variance; $\alpha \leq 4$: no kurtosis

Prerequisites for Today (2/19): Tails on Log-Log Axes



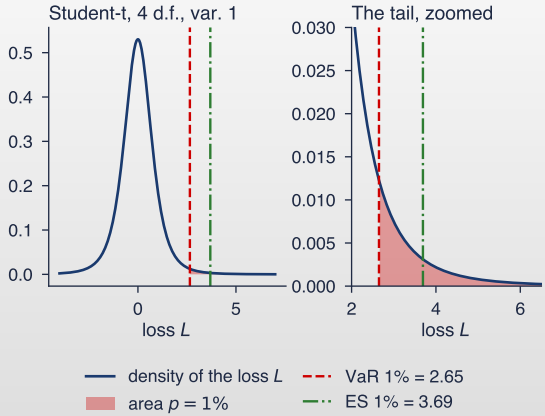
- $\bar{F}(x) = P(|L| > x)$ for four laws with variance 1
  - Student-t with ν d.f. (degrees of freedom): a bell-shaped law with a power tail, $\alpha = \nu$
- Power tail: a straight line

$$\ln \bar{F}(x) \approx \ln C - \alpha \ln x$$

  - the slope is $-\alpha$; the black segment has slope -3
- Light tails bend downwards
  - Normal: $\bar{F}(x)$ falls like $e^{-x^2/2}$; Laplace: like e^{-cx}, with $c > 0$
  - at 10 standard deviations: Student-t (3 d.f.) 4×10^{-4}, Normal 1.5×10^{-23}

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Prerequisites for Today (3/19): VaR and ES



- VaR_p: the loss exceeded with probability p, i.e. the $(1-p)$ quantile of the loss

$$P(L > \mathrm{VaR}_p) = p$$

  - the red area beyond VaR has probability $p = 1\%$; today: VaR 1%, VaR 0.1%, ES 2.5%

- ES_p: the mean loss in that area

$$\mathrm{ES}_p = E[L \mid L > \mathrm{VaR}_p]$$

  - the centre of gravity of the red area: always $\mathrm{ES}_p \geq \mathrm{VaR}_p$

- Chart: Student-t with 4 d.f. and variance 1: VaR 1% = 2.65, ES 1% = 3.69

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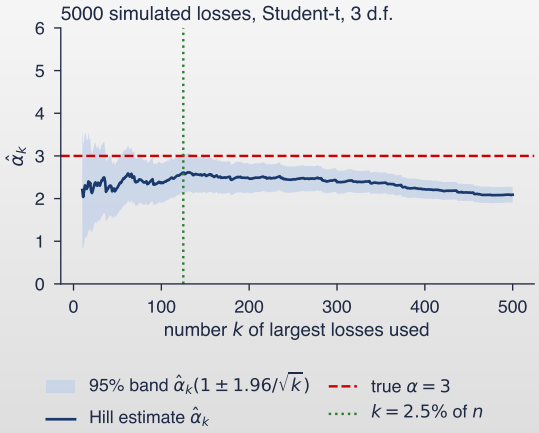
Prerequisites for Today (4/19): VaR, ES and Mean Excess of a Pareto Tail

- **Pareto tail**: $P(L > x) = p_0 \, (x/x_0)^{-\alpha}$ for $x \geq x_0$
  - x_0: the level where the power law starts; $p_0 = P(L > x_0)$
- Solving $p_0 (x/x_0)^{-\alpha} = p$ for x: $\mathrm{VaR}_p = x_0 \, (p_0/p)^{1/\alpha}$, for $p < p_0$
  - $\mathrm{ES}_p = \mathrm{VaR}_p \, \alpha/(\alpha - 1)$ for $\alpha > 1$
  - mean excess $e(u) = E[L - u \mid L > u] = u/(\alpha - 1)$: the average loss beyond u, rising with u

Prerequisites for Today (5/19): the Hill Estimator

- Order the losses $L_{(1)} \geq L_{(2)} \geq \cdots \geq L_{(n)}$ and use the k largest
  - $L_{(i)}$: the i-th largest loss; n: the number of days; $L_{(k+1)}$ plays the role of a threshold
- **Hill estimator** Hill (1975): $\hat{\alpha}_k = \left[\frac{1}{k} \sum_{i=1}^{k} \ln \frac{L_{(i)}}{L_{(k+1)}} \right]^{-1}$
  - the inverse of the average log ratio between the k largest losses and the threshold
  - the hat marks an estimate; a heavier tail gives larger ratios, so a smaller $\hat{\alpha}$

Prerequisites for Today (6/19): a Hill Plot



- **Hill plot**: $\hat{\alpha}_k$ against k
  - 5 000 simulated Student-t losses with 3 d.f.: the true tail index is $\alpha = 3$
- Small k: few losses, a wide band
  - the band narrows as k grows (its standard error: next slide)
- Large k: losses from the centre of the law enter the average
  - the estimate drifts away from 3: a bias
- Choose k where the curve is flat; today: $k = 2.5\%$ of n
  - here: $\hat{\alpha} = 2.61$, and the 95% band still contains 3

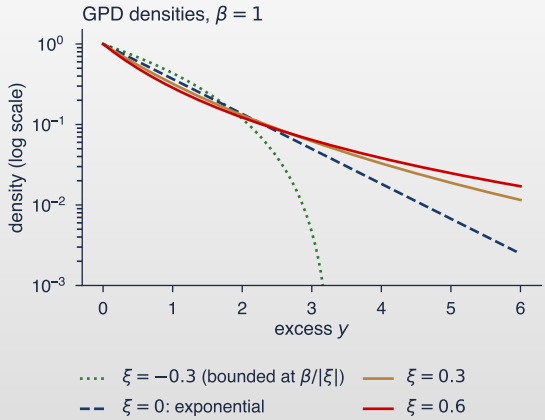
Prerequisites for Today (7/19): Precision of Hill and the Hill Quantile

- i.i.d. losses (independent, identically distributed): $\mathrm{SE} \approx \hat{\alpha}/\sqrt{k}$
  - SE: the standard error; 95% CI (confidence interval): $\hat{\alpha}(1 \pm 1.96/\sqrt{k})$
- **Moving-block bootstrap**: resample blocks of 20 consecutive days, recompute $\hat{\alpha}$ many times
  - the standard deviation of these estimates is an SE that keeps the clusters of large losses
- Comparing two estimates: $z = (\hat{\alpha}_1 - \hat{\alpha}_2)/\sqrt{\mathrm{SE}_1^2 + \mathrm{SE}_2^2}$
  - $|z| > 1.96$: a significant difference at 5%
- **Hill quantile** Weissman (1978): $\hat{x}_p = L_{(k+1)}\,(k/(np))^{1/\hat{\alpha}}$
  - the loss exceeded with probability p: the threshold, extrapolated with the power law

Prerequisites for Today (8/19): Peaks over Threshold

- **POT** (peaks over threshold): keep the losses above a high threshold u
  - the excesses $Y = L - u$ follow a **GPD** (generalised Pareto distribution) Pickands (1975), Balkema & de Haan (1974)
- $P(Y \leq y) = 1 - (1 + \xi y/\beta)^{-1/\xi}$, $y \geq 0$
  - shape ξ: the heaviness of the tail; $\xi > 0$: power tail with $\alpha = 1/\xi$
  - scale $\beta > 0$: the typical size of an excess
- Mean excess $e(v)$ above u for a GPD: $e(v) = (\beta + \xi(v - u))/(1 - \xi)$, $v \geq u$
  - a rising line in the threshold v; at $v = u$: $\beta/(1 - \xi)$, finite only for $\xi < 1$
- Threshold here: $u =$ the 90% quantile of the losses (the 10% largest losses, as in McNeil & Frey (2000))

Prerequisites for Today (9/19): the GPD Density



- Density of the excess: the derivative of the GPD CDF

$$g(y) = \frac{1}{\beta}\left(1 + \frac{\xi y}{\beta}\right)^{-1/\xi - 1}, \quad y \geq 0$$

  - $\xi = 0$: the limit $g(y) = e^{-y/\beta}/\beta$, the exponential law
- $\xi > 0$: the density falls like a power, $\alpha = 1/\xi$; the larger ξ, the heavier the tail
- $\xi < 0$: the excess is bounded, $y \leq \beta/|\xi|$

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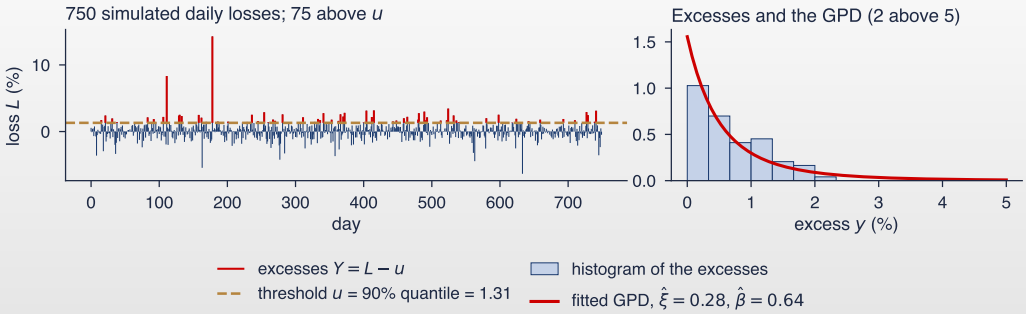
Prerequisites for Today (10/19): Maximum Likelihood for EVT

- **EVT** (extreme value theory): the statistics of the largest losses; POT is one of its methods
- **Maximum likelihood** (ML): the parameters that make the observed excesses most probable

$$\ell(\xi, \beta) = -N_u \ln \beta - \left(1 + \frac{1}{\xi}\right) \sum_{i=1}^{N_u} \ln\left(1 + \frac{\xi y_i}{\beta}\right)$$

  - ℓ: the log-likelihood, the sum of $\ln g(y_i)$ over the N_u excesses y_i
  - $(\hat{\xi}, \hat{\beta})$ maximise ℓ; they are found numerically
  - SE of $\hat{\xi}$: from the curvature of ℓ at its maximum; a flat ℓ means a large SE
  - 95% interval for ξ: $\hat{\xi} \pm 1.96\,\mathrm{SE}(\hat{\xi})$

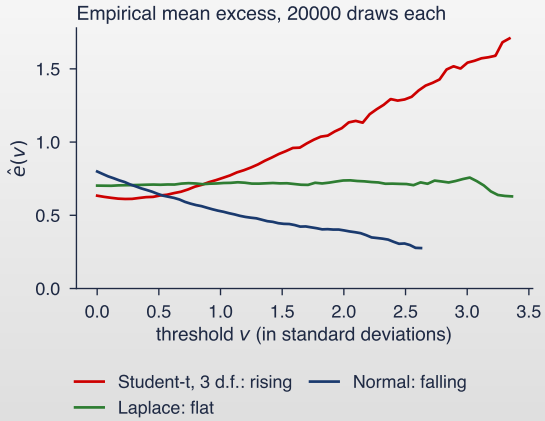
Prerequisites for Today (11/19): Peaks over Threshold on a Chart



- Left: 750 simulated daily losses; u is their 90% quantile, so 75 losses exceed it (red)
- Right: the excesses $Y = L - u$ and the GPD fitted by maximum likelihood, $\hat{\xi} = 0.28$, $\hat{\beta} = 0.64$

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Prerequisites for Today (12/19): the Empirical Mean Excess



- **Empirical mean excess** at a threshold v

$$\hat{e}(v) = \frac{\sum_t (L_t - v)\, 1\{L_t > v\}}{\sum_t 1\{L_t > v\}}$$

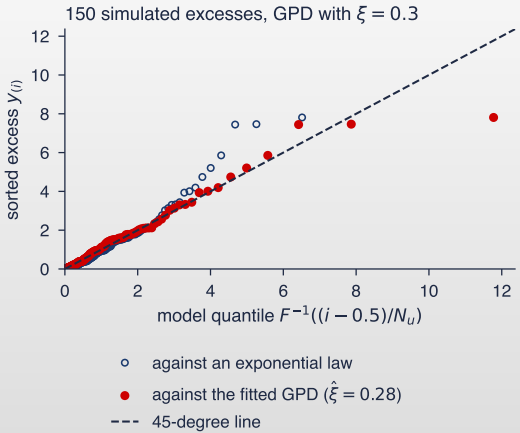
  - $1\{L_t > v\} = 1$ if the loss exceeds v, 0 otherwise: $\hat{e}(v)$ is the average of the excesses over v
- Rising line: a power tail (GPD with $\xi > 0$, slope $\xi/(1-\xi)$)
- Flat: an exponential tail ($\xi = 0$); falling: a tail lighter than exponential (Normal)
- Use: choose u where $\hat{e}(v)$ becomes roughly linear

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Prerequisites for Today (13/19): VaR and ES by POT

- n days, N_u losses above u; valid for $p < N_u/n$ McNeil & Frey (2000)
  - N_u/n: the share of days above the threshold; the GPD says nothing below u
- $\mathrm{VaR}_p = u + \frac{\beta}{\xi}\left[\left(\frac{np}{N_u}\right)^{-\xi} - 1\right]$
  - the threshold plus the GPD quantile of the excess; the exponent is $-\xi$
- $\mathrm{ES}_p = \frac{\mathrm{VaR}_p + \beta - \xi u}{1 - \xi}$, for $\xi < 1$
  - VaR plus the mean excess beyond VaR
- Two benchmarks
  - historical: the empirical $(1 - p)$ quantile of the losses and the mean of the losses beyond it
  - Normal: $\hat{\mu}_L + \hat{\sigma} z_{\mathbf{1}-p}$ and $\hat{\mu}_L + \hat{\sigma}\varphi(z_{\mathbf{1}-p})/p$, with $z_{\mathbf{1}-p}$ the $N(0,1)$ quantile and φ its density

Prerequisites for Today (14/19): QQ Plots



- **QQ plot**: the sorted data against the model quantiles

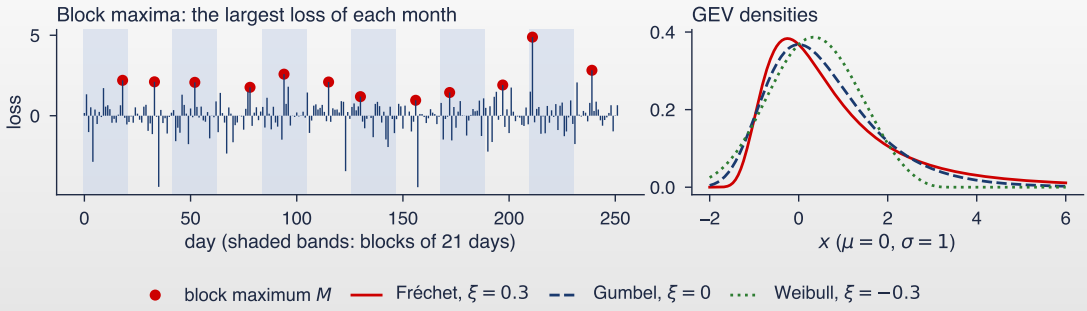
$$\left(F^{-1}\left(\frac{i-0.5}{N_u}\right),\ y_{(i)}\right), \quad i = 1, \ldots, N_u$$

  - $y_{(i)}$: the i-th smallest excess; F: the fitted CDF
- Good model: the points follow the 45-degree line (red: the fitted GPD)
- Points above the line on the right: the data have a heavier tail than the model (blue: the exponential law)
- The last few points always scatter: the largest values are very variable

Prerequisites for Today (15/19): Block Maxima and the GEV

- M = the largest daily loss in a block (a month, a quarter)
  - for long blocks M follows a **GEV** (generalised extreme value) law Fisher & Tippett (1928), Gnedenko (1943)
- $P(M \le x) = \exp\{-[1+\xi(x-\mu)/\sigma]^{-\mathbf{1}/\xi}\}$
  - μ: location (typical maximum); $\sigma > 0$: scale (spread); ξ: shape of the tail
  - $\xi = 0$: Gumbel, $\exp\{-e^{-(x-\mu)/\sigma}\}$
  - $\xi > 0$ Fréchet (heavy tails), $\xi = 0$ Gumbel (light), $\xi < 0$ Weibull (bounded)
- GEV fitted by maximum likelihood to the block maxima $M_{\mathbf{1}}, \ldots, M_m$, as the GPD to the excesses
  - if the 95% interval for ξ excludes 0, the tail is Fréchet

Prerequisites for Today (16/19): Block Maxima on a Chart



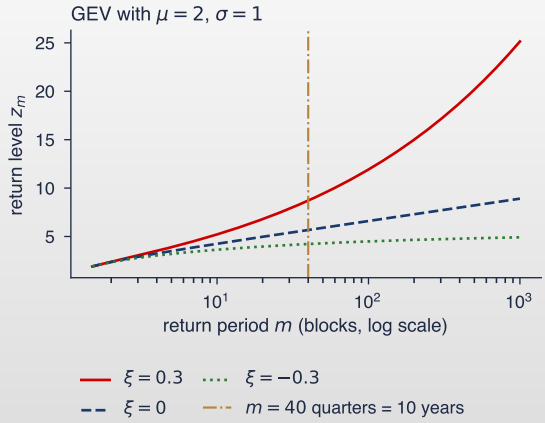
- Left: one year of simulated daily losses in 12 blocks of 21 days; the red points are the 12 block maxima M
- Right: the three GEV types; Fréchet ($\xi > 0$) has the heaviest right tail, the usual case for financial losses

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Prerequisites for Today (17/19): Return Levels and Checks

- **Return level** for m blocks: the level exceeded on average once in m blocks
  - $z_m = \mu + \frac{\sigma}{\xi}\left[(-\ln(1-1/m))^{-\xi} - 1\right]$: the GEV quantile of order $1 - 1/m$
  - 10 years = 120 months = 40 quarters; return period of a level x: $1/P(M > x)$ blocks
- **Out-of-sample check**: estimate VaR on a first period, count the days of a second period with a loss above it
  - if VaR is right, the count $X \sim \mathrm{Binomial}(n, p)$, with mean np
  - two-sided p-value $2\min\{P(X \leq x), P(X \geq x)\}$, with x the observed count; small p-value: VaR rejected
- Quick checks: VaR 0.1% $>$ VaR 1%; ES 2.5% $>$ VaR 2.5%; a GPD with $\xi < 1/2$ has a finite variance

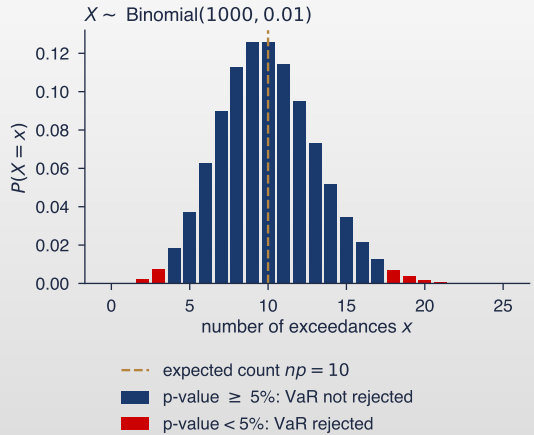
Prerequisites for Today (18/19): Return Levels on a Chart



- The return level z_m of the previous slide, against m
  - Gumbel ($\xi = 0$): $z_m = \mu - \sigma \ln(-\ln(1 - 1/m))$
- $\xi > 0$ (red): z_m grows ever faster with m; $\xi < 0$ (green): it approaches the upper bound $\mu - \sigma/\xi$
- Beyond the length of the sample the curve is an extrapolation: its uncertainty grows with m

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Prerequisites for Today (19/19): Counting VaR Exceedances



- VaR 1% on $n = 1\,000$ test days: the number of exceedances

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, \ldots, n$$

  - if VaR is right: mean $np = 10$, standard deviation $\sqrt{np(1-p)} \approx 3.1$
- Blue: counts with a two-sided p-value of at least 5% (here 4 to 17): VaR not rejected
- Red: VaR rejected
  - too few exceedances: VaR too high (too prudent)
  - too many exceedances: VaR too low (risk underestimated)

A1: Hill from Eight Losses [Solved]

Task

- The eight largest daily losses (%) of a stock, in $n = 2500$ days: $12.4, 9.8, 8.1, 7.5, 6.9, 6.2, 5.8, 5.5$.
- 1. Compute the Hill estimate with $k = 7$.
- 2. Compute its standard error and the 95% interval.
- 3. Compute the Hill quantile for $p = 0.1\%$.
- Report: $\hat{\alpha}$, SE, the interval, VaR 0.1% and one sentence on the moments.

Solution

- 1. $\ln(L_{(i)}/5.5)$: $0.813, 0.578, 0.387, 0.310, 0.227, 0.120, 0.053$; mean 0.355; $\hat{\alpha} = 2.81$
- 2. $\mathrm{SE} = 2.81/\sqrt{7} = 1.06$; interval $[0.73, 4.90]$: very wide with $k = 7$
- 3. $k/(np) = 7/2.5 = 2.800$; $2.800^{1/2.81} = 1.44$; $\hat{x}_{0.001} = 5.5 \times 1.44 = 7.93\%$
- $\hat{\alpha} \approx 2.8$: variance finite, kurtosis infinite; but the interval also allows $\alpha < 2$.

Back Exercise list

A2: Hill for a Crypto Asset [Proposed]

Solution discussed in class

Task

- The nine largest daily losses (%) in $n = 3000$ days: $15.2, 11.0, 9.4, 8.0, 7.7, 7.1, 6.6, 6.3, 6.0$. Model: A1.
- 1. Compute the Hill estimate with $k = 8$, its standard error and the 95% interval.
- 2. Compute the Hill quantile for $p = 0.1\%$.
- 3. Explain why the interval is so wide and what would make it narrower.
- Report: $\hat{\alpha}$, SE, the interval, VaR 0.1% and one sentence.

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A3: A Pareto Tail [Solved]

Task

- Daily losses: $P(L > x) = 2\% \times (x/2.5)^{-3}$ for $x \geq 2.5\%$.
- 1. Compute $P(L > 5\%)$ and $P(L > 10\%)$.
- 2. Compute VaR 1%, VaR 0.1% and ES 1%.
- 3. Compute the mean excess at $u = 4\%$.
- 4. Compare $P(L > 5\%)$ and $P(L > 10\%)$ with a Normal law $N(0, \sigma^2)$ that has the same $P(L > 2.5\%) = 2\%$.
- Report: four probabilities, three risk numbers, $e(4)$ and one sentence.

Solution

- 1. $2\% \times 2^{-3} = 0.250\%$; $2\% \times 4^{-3} = 0.031\%$
- 2. $\mathrm{VaR}_p = 2.5\,(2\%/p)^{1/3}$: VaR 1% $= 3.15\%$, VaR 0.1% $= 6.79\%$; ES 1% $= 3.15 \times 3/2 = 4.72\%$
- 3. $e(u) = u/(\alpha - 1) = 4/2 = 2.00\%$
- 4. $\sigma = 2.5/z_{0.98} = 1.22$; Normal: $P(L > 5) = 2.0 \times 10^{-3}\%$, $P(L > 10) = 1.1 \times 10^{-14}\%$: the Normal tail vanishes

Back Exercise list

A4: A Heavier Pareto Tail [Proposed]

Solution discussed in class

Task

- $P(L > x) = 1.5\% \times (x/3)^{-2.5}$ for $x \geq 3\%$. Model: A3.
- 1. Compute $P(L > 6\%)$ and $P(L > 12\%)$.
- 2. Compute VaR 0.5%, VaR 0.1%, ES 0.5% and the mean excess at $u = 5\%$.
- 3. Say which moments of L exist.
- Report: two probabilities, four numbers and one sentence.

Back Exercise list

A5: A GEV Return Level [Solved]

Task

- Monthly maxima of daily losses (%) follow a GEV with $\xi = 0.25$, $\mu = 1.5$, $\sigma = 0.8$.
- 1. Compute the probability that the largest daily loss of a month exceeds 6%.
- 2. Compute the return period of a 6% loss in years.
- 3. Compute the 10-year return level.
- 4. Name the type of the GEV and its tail index.
- Report: one probability, one period, one level and one sentence.

Solution

- 1. $1 + 0.25(6 - 1.5)/0.8 = 2.406$;
  $P = 1 - \exp\{-2.406^{-4}\} = 2.94\%$
- 2. $1/P = 34$ months, i.e. about 2.8 years
- 3. $m = 120$: $-\ln(1 - 1/120) = 0.00837$; $0.00837^{-0.25} = 3.306$;
  $z = 1.5 + (0.8/0.25)(3.306 - 1) = 8.88\%$
- 4. $\xi > 0$: Fréchet, tail index $1/\xi = 4$.

Back Exercise list

A6: The Same Question with a Gumbel Law [Proposed]

Solution discussed in class

Task

- Monthly maxima follow a Gumbel law ($\xi = 0$) with $\mu = 1.5$, $\sigma = 0.8$. Model: A5.
- 1. Compute $P(M > 6\%)$ and the return period of 6% in years.
- 2. Compute the 10-year return level ($m = 120$) with the Gumbel formula of Prerequisites for Today (18/19).
- 3. Compare with A5: what does $\xi = 0.25$ change?
- Report: one probability, one period, one level and one sentence.

Back Exercise list

A7: POT Risk Measures, Step by Step [Solved]

Task

- $n = 5000$ daily losses; $u = 2\%$; $N_u = 250$ losses above u; GPD fit: $\xi = 0.2$, $\beta = 0.8$.
- 1. Compute VaR 1% and VaR 0.1%.
- 2. Compute VaR 2.5% and ES 2.5%.
- 3. Compute the mean excess at u and the tail index.
- Report: four risk numbers, $e(u)$, α and one sentence.

Solution

- 1. $\beta/\xi = 4.00$; $p = 1\%$: $np/N_u = 0.200$, $0.200^{-0.2} = 1.380$, VaR $= 2 + 4(1.380 - 1) = 3.52\%$; $p = 0.1\%$: 2.187, VaR $= 6.75\%$
- 2. $np/N_u = 0.500$, VaR 2.5% $= 2.59\%$; ES $= (2.59 + 0.8 - 0.4)/0.8 = 3.74\%$
- 3. $e(u) = \beta/(1 - \xi) = 1.00\%$; $\alpha = 1/\xi = 5$: a heavy tail, yet with a finite kurtosis.

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A8: POT for a Crypto Asset [Proposed]

Solution discussed in class

Task

- $n = 4000$; $u = 3.1\%$; $N_u = 200$; GPD fit: $\xi = 0.3$, $\beta = 1.5$. Model: A7.
- 1. Compute VaR 1%, VaR 0.1% and ES 2.5%.
- 2. Compute the ratio ES 1% / VaR 1% and compare it with $1/(1-\xi)$.
- 3. Say whether the formulas can be used for VaR 10%, and why.
- Report: three risk numbers, one ratio and two sentences.

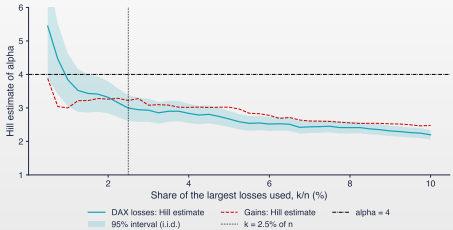
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B1: How Heavy Is the Tail of DAX Losses? [Solved]

- **Question**: what is the tail index of daily DAX losses, how precise is it, and does the kurtosis of DAX returns exist?
- **Inputs**: DAX closes, 1990–2026, daily losses $L_t = -r_t$ in %
- **Tasks**
  1. Draw the Hill plot of the losses for k between 0.5% and 10% of n, with the i.i.d. 95% band.
  2. Compute $\hat{\alpha}$ at $k = 2.5\%$ of n with the i.i.d. SE, the 95% interval and the moving-block bootstrap SE (blocks of 20 days, 500 samples).
  3. Compute the Hill estimate of the gains with the same k.
  4. Interpretation: is $\alpha > 4$ (a finite kurtosis) compatible with the data?
- **Report**: the chart, four numbers and one sentence
- Notebook: section B1

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B1: Solution [Solved]



- $n = 9\,288$, $k = 232$, $L_{(k+\mathbf{1})} = 2.86\%$: $\hat{\alpha} = 3.00$
  - SE i.i.d. 0.20, interval $[2.61, 3.38]$, SE block 0.20; gains: 3.22
- Between $k = 1\%$ and 5% of n the estimate falls slowly, from 3.84 to 2.67; the gains have a lighter tail
- Interpretation: $z = (\hat{\alpha} - 4)/\mathrm{SE}_{\mathbf{block}} = -5.0$, so $\alpha = 4$ is rejected
  - the kurtosis of DAX returns does not exist: the sample kurtosis (6.0) is not a stable number

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B2: BET, Bitcoin and Two BVB Stocks [Proposed]

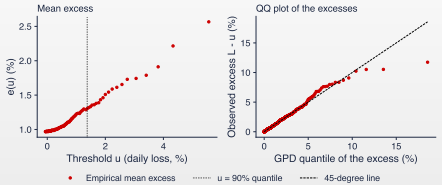
- **Question**: which of the BET, Bitcoin, OMV Petrom and Banca Transilvania has the heaviest tail of losses? Model: B1.
- **Inputs**: BET since 1997, Bitcoin since 2014, SNP and TLV since 2010 (TLV without 30–31 May 2016), daily losses in %
- **Tasks**
  1. Compute $\hat{\alpha}$ at $k = 2.5\%$ of n for each series, with the 95% interval and the block-bootstrap SE.
  2. Add the estimates at $k = 1\%$ and 5% of n.
  3. Test the difference between the BET and Bitcoin with the z statistic of Prerequisites for Today (7/19), using the block-bootstrap SEs.
  4. Interpretation: is Bitcoin's tail heavier than the BET's once the scale of the losses is set aside?
- **Report**: one table, one z and two sentences
- Notebook: section B2

B3: POT for the BET [Solved]

- **Question**: how large are VaR 1%, ES 2.5% and VaR 0.1% of daily BET losses by EVT, compared with historical simulation and the Normal distribution?
- **Inputs**: BET closes since 1997, daily losses in %; u = the 90% quantile of the losses
- **Tasks**
  1. Draw the empirical mean excess function and mark u.
  2. Fit a GPD to the excesses over u by maximum likelihood; report $\hat{\xi}$ with its SE and $\hat{\beta}$.
  3. Draw the QQ plot of the excesses against the fitted GPD.
  4. Compute VaR 1%, ES 2.5% and VaR 0.1% by EVT, by historical simulation and by the Normal distribution.
  5. Interpretation: which method would you use for the capital of a bank holding the BET, and why?
- **Report**: the two charts, a table of nine numbers and two sentences
- Notebook: section B3

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B3: Solution [Solved]



(%)	EVT	historical	Normal
VaR 1%	4.44	4.33	3.39
ES 2.5%	4.81	4.80	3.41
VaR 0.1%	9.56	9.64	4.52

- $n = 7\,243$, $u = 1.375\%$, $N_u = 725$; $\hat{\xi} = 0.223$ (SE 0.047), $\hat{\beta} = 1.018$
  - the mean excess rises linearly above u; the QQ plot is on the line except the largest excesses
- Interpretation: EVT, which agrees with the data at 1% and gives a smooth VaR 0.1%; the Normal VaR 0.1% is less than half

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B4: Does the Threshold Matter? Bitcoin [Proposed]

- **Question**: how much do $\hat{\xi}$ and the EVT risk measures of Bitcoin change with the threshold? Model: B3.
- **Inputs**: Bitcoin since 2014, daily losses in %; u at the 85%, 90% and 95% quantiles
- **Tasks**
  1. For each threshold, fit the GPD and report u, N_u, $\hat{\xi}$ with its SE and $\hat{\beta}$.
  2. Compute VaR 1%, ES 2.5% and VaR 0.1% for each threshold, and by historical simulation.
  3. Interpretation: which quantity is robust to the choice of u, $\hat{\xi}$ or the VaR?
- **Report**: one table and two sentences
- Notebook: section B4

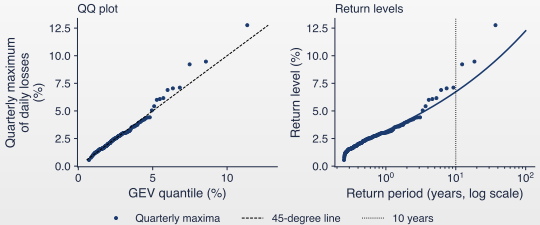
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B5: Quarterly Maxima of the S&P 500 [Solved]

- **Question**: how large is the daily loss that the S&P 500 exceeds once in 10 years, according to a GEV fit to quarterly maxima?
- **Inputs**: S&P 500 closes since 1990, the largest daily loss of each calendar quarter
- **Tasks**
  1. Build the quarterly maxima of the daily losses.
  2. Fit a GEV by maximum likelihood; report $\hat{\xi}$ with its 95% interval, $\hat{\mu}$ and $\hat{\sigma}$.
  3. Draw the QQ plot and the return level plot.
  4. Compute the 10-year return level (40 quarters) and count the quarters above it.
  5. Interpretation: is the S&P 500 tail Fréchet or Gumbel?
- **Report**: three parameters, one level, the charts and one sentence
- Notebook: section B5

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B5: Solution [Solved]



- 147 quarters: $\hat{\xi} = 0.21$, 95% interval $[0.09, 0.32]$; $\hat{\mu} = 1.97\%$, $\hat{\sigma} = 0.86\%$
- 10-year return level: 6.7%; exceeded in 6 of 147 quarters (3.7 expected in 36.8 years); largest loss 12.8%
- Interpretation: $\xi = 0$ lies outside the interval: Fréchet, a heavy tail with $\alpha = 1/\hat{\xi} = 4.8$

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B6: Does VaR 1% Survive a New Period? [Proposed]

- **Question**: estimated until 2014, is the VaR 1% of DAX and BET losses exceeded on about 1% of the days of 2015–2026? Models: B3, B5.
- **Inputs**: DAX since 1990, BET since 1997, daily losses in %; estimation until 31 December 2014, test afterwards
- **Tasks**
  1. Estimate VaR 1% on the first period by EVT (POT, u at the 90% quantile), historical simulation and the Normal distribution.
  2. Count the test days with a loss above each VaR and compare with np.
  3. Compute the two-sided binomial p-value of each count.
  4. Interpretation: why can all three methods fail in the same direction?
- **Report**: a table with six counts and p-values, and two sentences
- Notebook: section B6

C1: Has the Tail of Bitcoin Changed? [Proposed]

- **Question**: is the tail index of Bitcoin losses different in 2014–2017, 2018–2021 and 2022–2026?
- **Inputs**: Bitcoin daily losses since 2014, split into three periods; models: B1, B2
- **Tasks**
  1. Compute the Hill $\hat{\alpha}$ at $k = 2.5\%$ of n in each period, with the block-bootstrap SE and a 95% interval.
  2. Say whether any two periods differ significantly.
  3. Propose one change to the method that would give a sharper answer.
  4. Interpretation: what can a sample of about 1 500 days say about a tail index?
- **Report**: a table, one sentence per comparison and a plan for a project
- Notebook: section C1

C2: Audit an AI Answer [Proposed]

- A student asked an AI assistant to estimate the tail of daily S&P 500 losses since 1990 by POT. The answer:
- `(a) With u = 1.163%, N_u = 924 of n = 9 240, the GPD gives xi = 0.158, so the tail index is alpha = 0.158 and the variance is infinite.`
- `(b) VaR 1% = u + (beta/xi) [(n p / N_u)^xi - 1] = −0.32%.`
- `(c) ES 2.5% = VaR 2.5% x (1 - xi) = 1.98%.`
- `(d) The same formula gives a reliable VaR 20%.`
- `(e) Above u, the mean excess function of the fitted GPD is a straight line in the threshold.`
- Tasks
  - 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
  - 2. Report: a list of five verdicts with one line of justification each.

What You Should Take from Today

- Daily losses have power tails with α between 2.5 and 3.5: finite variance, no finite kurtosis
- Hill: fix k in a flat region of the Hill plot; report a standard error that allows for clustering
- POT: u at a high quantile, GPD for the excesses, VaR and ES by closed formulas; check with mean excess and QQ plots
- Block maxima: GEV, Fréchet for financial losses; return levels beyond the sample are extrapolation
- An AI answer is a draft: check $\alpha = 1/\xi$, the sign of the exponent, ES $\geq$ VaR and the range $p < N_u/n$

After the Seminar

- Lecture 5 develops each topic of today: crashes, regular variation, Hill, mean excess, GEV, GPD, EVT VaR and ES
- C1 can grow into a team project: GARCH-standardised losses, POT with a common threshold, more crypto assets
- Reading: Franke, Härdle & Hafner (2019), Ch. 18; exercises in Borak, Härdle & López-Cabrera (2013), Ch. 18; McNeil, Frey & Embrechts (2015), Ch. 5

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